

Notation Reference

Notation Reference I

This paper intentionally minimizes mathematical notation to maximize accessibility. Where notation is used, it follows the Cognitive Integrity Framework (CIF) formal specification defined in Part 1 of this series.

Minimal Notation Used I

Symbol	Meaning	Plain Language
	Trust decay factor	"Delegated trust decreases by this factor at each step"
n	Agent count	"Number of agents in the system"
f	Byzantine agents	"Maximum number of malicious agents tolerated"

Trust Decay Explanation I

When we write $\gamma = 0.9$, this means:

- ▶ Direct trust: 100% of assigned value
- ▶ One delegation: 90% of source trust
- ▶ Two delegations: 81% of source trust
- ▶ Three delegations: 73% of source trust

A lower γ (e.g., 0.85) means faster decay, providing more security but limiting delegation utility.

Byzantine Tolerance Explanation I

When we say $n \geq 3f + 1$:

- ▶ To tolerate 1 malicious agent, need at least 4 agents
- ▶ To tolerate 2 malicious agents, need at least 7 agents
- ▶ To tolerate 3 malicious agents, need at least 10 agents

Full Notation Reference I

For complete formal definitions of all CIF notation, see Part 1 supplementary **S03** (`cogsec_multiagent_1_theory/manuscript/S03_notation.md` in this repository's `cognitive_integrity` program tree).

The Part 1 specification uses on the order of 100 symbols covering:

- ▶ Agent cognitive state
- ▶ Trust calculus operations
- ▶ Defense mechanism parameters
- ▶ Consensus and coordination
- ▶ Information-theoretic bounds